Akoizo Original Practice - 2027 Rotation Form 1

Division B | Crime Busters | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) Which solid turns brown iodine colorless?

Unit: Identifying white powders.
Worked-case data: An unknown white powder is insoluble in water, fizzes strongly with HCl, and leaves iodine yellow-brown. Identify it.
Separate investigation observations:
Appearance: Fine, pure white powder that squeaks when pressed.
Water: Does not dissolve; forms a cloudy suspension.
HCl: No fizzing.
Iodine: Turns blue-black.

A. Cornstarch

B. Table salt

C. Gypsum

D. Vitamin C

Answer: _____

2. (2 points) Which listed metal does not react with hydrochloric acid?

Unit: Metals and liquids.
Worked-case data: A silvery metal is not magnetic, reacts vigorously with HCl, and has a density of about 1.7 g/cm^3. Identify it.
Separate investigation observations:
Liquid color: Clear and colorless.
Odor: Little odor.
pH paper: About pH 6, close to neutral.
With yeast: Vigorous foaming and bubbling.

A. Iron

B. Copper

C. Zinc

D. Magnesium

Answer: _____

3. (2 points) A human hair usually has a medullary index of:

Unit: Hair and fibers.
Worked-case data: A hair has a continuous medulla about 60% of the hair’s width and scales shaped like petals. What is its source?
Separate investigation observations:
Hair A: Medulla fragmented; index about 0.2.
Hair B: Wide, continuous ladder-like medulla; flattened scales.
Fiber: Burns quickly with afterglow and soft gray ash.

A. Greater than one-half

B. Exactly one

C. Zero always

D. Less than about one-third

Answer: _____

4. (2 points) Which plastics float in water?

Unit: Recyclable plastics.
Worked-case data: A plastic fragment sinks in water, floats in a 1.10 g/cm^3 salt solution, and burns with a sooty flame. Identify it.
Separate investigation observations:
Water: Sinks.
1.10 g/cm^3 solution: Sinks.
1.25 g/cm^3 solution: Floats.
Appearance: Very clear, rigid, and shatter-resistant.

A. All plastics

B. PP, LDPE, and HDPE

C. PVC and PETE

D. PS and PMMA

Answer: _____

5. (2 points) Why is the starting line drawn in pencil?

Unit: Paper chromatography.
Worked-case data: Ransom-note ink separates into a blue spot at 2.4 cm and a yellow spot at 4.8 cm, with the solvent front at 6.0 cm. Suspect A's pen gives Rf values 0.40 and 0.80; suspect B's gives 0.40 and 0.65. Which pen matches?
Separate investigation observations:
Note ink: Three bands: Rf 0.15, 0.45, 0.70.
Suspect 1 pen: Two bands: Rf 0.45, 0.70.
Suspect 2 pen: Three bands: Rf 0.15, 0.45, 0.70.
Suspect 3 pen: Three bands: Rf 0.20, 0.50, 0.90.

A. Pencil speeds up the solvent

B. Ink is required for Rf

C. Pencil changes Rf values

D. Pencil graphite does not dissolve and separate

Answer: _____

6. (2 points) Which fingerprint pattern has no delta?

Unit: Fingerprints, DNA, prints, soil, and spatter.
Worked-case data: A scene DNA sample shows peaks at 12 and 15 repeats at location 1, and 8 and 8 at location 2. Suspect X: 12/15 and 8/10. Suspect Y: 12/15 and 8/8. Which suspect matches?
Separate investigation observations:
Scene print: A whorl with two deltas.
Suspect A: Right index finger: loop.
Suspect B: Right index finger: whorl with matching ridge endings.
Suspect C: Right index finger: arch.

A. Double loop

B. Arch

C. Loop

D. Whorl

Answer: _____

7. (2 points) An analysis must explain the culprit and also:

Unit: Writing the crime scene analysis.
Worked-case data: Evidence: the scene powder is baking soda; suspect A has baking soda, B has cornstarch, C has baking soda. Note ink matches only C's pen. A dog hair was found; A and C own dogs, B owns a cat. Who is the culprit?
Separate investigation observations:
Powder: Calcium carbonate: found with suspects 1 and 3.
Plastic: HDPE fragment: suspects 1, 2, and 3 all use HDPE jugs.
Chromatography: Note ink matches only suspect 3's pen.
Fingerprint: Scene loop matches suspect 3's minutiae.

A. Only the strongest evidence

B. Each suspect's alibi only

C. The price of each test

D. Why the other suspects were not chosen

Answer: _____

8. (2 points) Which solid fizzes in plain water without added acid?

Unit: Identifying white powders.
Worked-case data: An unknown white powder is insoluble in water, fizzes strongly with HCl, and leaves iodine yellow-brown. Identify it.
Separate investigation observations:
Appearance: Fine, pure white powder that squeaks when pressed.
Water: Does not dissolve; forms a cloudy suspension.
HCl: No fizzing.
Iodine: Turns blue-black.

A. Baking soda

B. Calcium carbonate

C. Sugar

D. Alka-Seltzer

Answer: _____

9. (2 points) Which liquid is basic with a strong, pungent odor?

Unit: Metals and liquids.
Worked-case data: A silvery metal is not magnetic, reacts vigorously with HCl, and has a density of about 1.7 g/cm^3. Identify it.
Separate investigation observations:
Liquid color: Clear and colorless.
Odor: Little odor.
pH paper: About pH 6, close to neutral.
With yeast: Vigorous foaming and bubbling.

A. Lemon juice

B. Household ammonia

C. Vinegar

D. Hydrogen peroxide

Answer: _____

10. (2 points) A fiber burns quickly, glows afterward, and leaves soft gray ash. It is:

Unit: Hair and fibers.
Worked-case data: A hair has a continuous medulla about 60% of the hair's width and scales shaped like petals. What is its source?
Separate investigation observations:
Hair A: Medulla fragmented; index about 0.2.
Hair B: Wide, continuous ladder-like medulla; flattened scales.
Fiber: Burns quickly with afterglow and soft gray ash.

A. Animal (such as wool)

B. Synthetic (such as nylon)

C. Mineral

D. Vegetable (such as cotton)

Answer: _____

11. (2 points) Identify the powder from the supplied reference comparison and distinguish observation from inference.

Synthetic complete case 1: crime-scene powder is water-soluble, produces gas with supplied dilute acid, and has the same test pattern as reference sodium bicarbonate. Scene fiber matches the provided cotton reference in microscopy and controlled test observations. A suspect's pen has two chromatography bands at 3 and 5 cm with a solvent front at 9 cm. Reference results are supplied to avoid any need for physical unknowns.

Answer: _____
_____
_____

12. (2 points) Why is gas production with acid alone insufficient to uniquely identify this powder?

Synthetic complete case 1: crime-scene powder is water-soluble, produces gas with supplied dilute acid, and has the same test pattern as reference sodium bicarbonate. Scene fiber matches the provided cotton reference in microscopy and controlled test observations. A suspect's pen has two chromatography bands at 3 and 5 cm with a solvent front at 9 cm. Reference results are supplied to avoid any need for physical unknowns.

Answer: _____
_____
_____

13. (2 points) Calculate the Rf value for the first listed band.

Synthetic complete case 1: crime-scene powder is water-soluble, produces gas with supplied dilute acid, and has the same test pattern as reference sodium bicarbonate. Scene fiber matches the provided cotton reference in microscopy and controlled test observations. A suspect's pen has two chromatography bands at 3 and 5 cm with a solvent front at 9 cm. Reference results are supplied to avoid any need for physical unknowns.

Answer: _____
_____
_____

14. (2 points) Calculate the Rf value for the second listed band.

Synthetic complete case 1: crime-scene powder is water-soluble, produces gas with supplied dilute acid, and has the same test pattern as reference sodium bicarbonate. Scene fiber matches the provided cotton reference in microscopy and controlled test observations. A suspect's pen has two chromatography bands at 3 and 5 cm with a solvent front at 9 cm. Reference results are supplied to avoid any need for physical unknowns.

Answer: _____
_____
_____

15. (2 points) Does a matching ink-band pattern identify one unique pen owner? Explain.

Synthetic complete case 1: crime-scene powder is water-soluble, produces gas with supplied dilute acid, and has the same test pattern as reference sodium bicarbonate. Scene fiber matches the provided cotton reference in microscopy and controlled test observations. A suspect's pen has two chromatography bands at 3 and 5 cm with a solvent front at 9 cm. Reference results are supplied to avoid any need for physical unknowns.

Answer: _____
_____
_____

16. (2 points) Classify the supplied cotton reference fiber as natural or synthetic and state its source category.

Synthetic complete case 1: crime-scene powder is water-soluble, produces gas with supplied dilute acid, and has the same test pattern as reference sodium bicarbonate. Scene fiber matches the provided cotton reference in microscopy and controlled test observations. A suspect's pen has two chromatography bands at 3 and 5 cm with a solvent front at 9 cm. Reference results are supplied to avoid any need for physical unknowns.

Answer: _____
_____
_____

17. (2 points) Explain why a human hair morphology match alone does not identify one person.

Answer: _____
_____
_____

18. (2 points) A suspect has a powder match but a non-matching fiber and ink. Explain how to report the mixed evidence.

Synthetic complete case 1: crime-scene powder is water-soluble, produces gas with supplied dilute acid, and has the same test pattern as reference sodium bicarbonate. Scene fiber matches the provided cotton reference in microscopy and controlled test observations. A suspect's pen has two chromatography bands at 3 and 5 cm with a solvent front at 9 cm. Reference results are supplied to avoid any need for physical unknowns.

Answer: _____
_____
_____

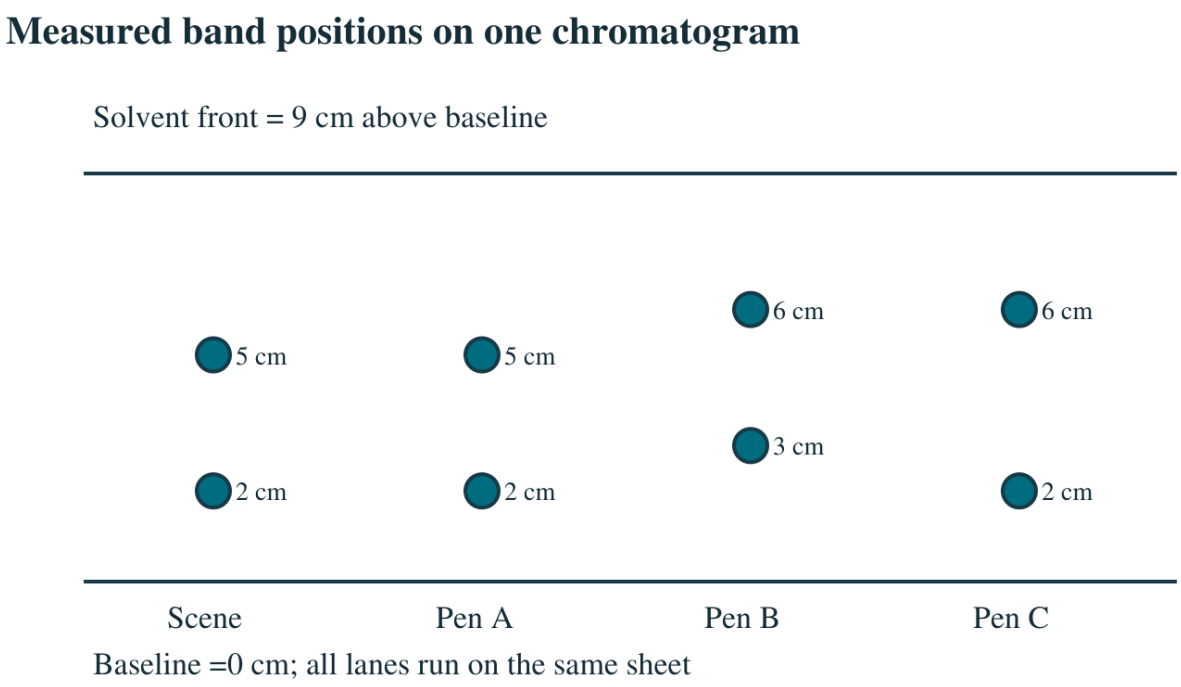
19. (2 points) Describe one control that helps distinguish contamination from a real sample result.

Answer: _____
_____
_____

20. (2 points) Why should a complete case report retain contradictory observations rather than omit them?

Answer: _____
_____
_____

21. (2 points) Use Figure F1. Calculate Rf for the lower scene band.



F1. Ink chromatography (original synthetic teaching figure)

Answer: _____
_____
_____

22. (2 points) Use Figure F1. Which candidate pen matches both scene band positions?

Refer to F1. Ink chromatography (original synthetic teaching figure) shown with the first question in this figure group.

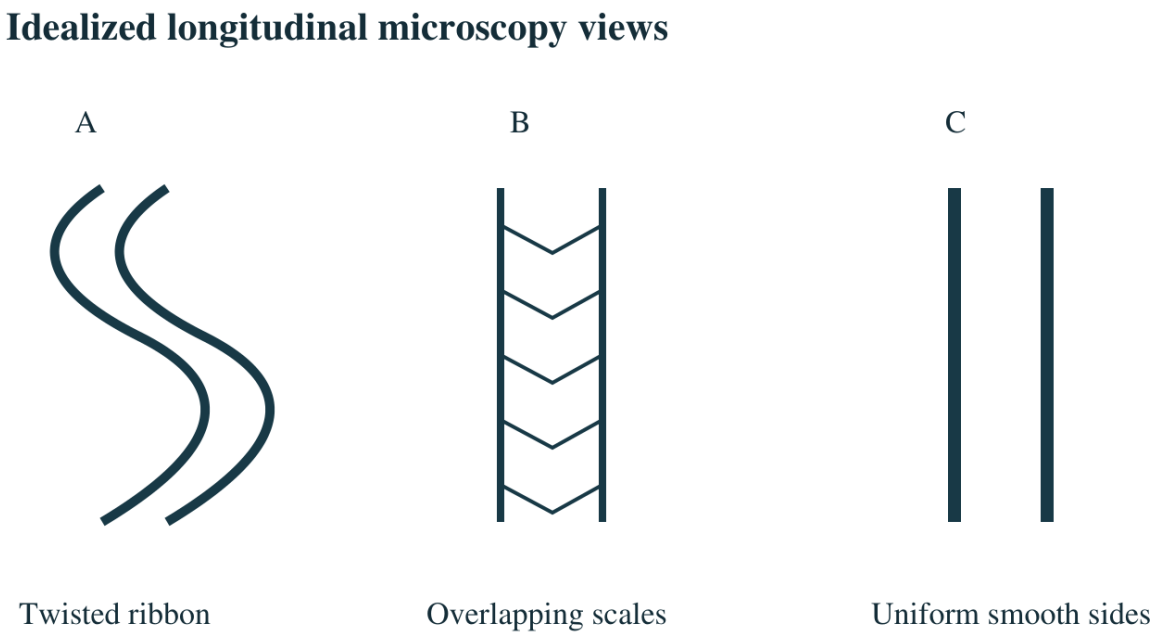
Answer: _____
_____
_____

23. (2 points) Use Figure F1. Does the match uniquely identify a writer? Explain.

Refer to F1. Ink chromatography (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

24. (2 points) Use Figure F2. Which panel is most consistent with the supplied cotton reference morphology?



F2. Fiber morphology sketches (original synthetic teaching figure)

Answer: _____
_____
_____

25. (2 points) Use Figure F2. Which panel is most consistent with wool-type cuticle morphology?

Refer to F2. Fiber morphology sketches (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

26. (2 points) Use Figure F2. Why is assigning one unique garment from panel C alone unjustified?

Refer to F2. Fiber morphology sketches (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

27. (2 points) Use Figure F3. Which supplied reference best matches U using the complete pattern?

Matched test conditions: supplied references and unknown

Sample	Water	Dilute acid	Iodine
Ref bicarbonate	Dissolves	Bubbles	No blue-black
Ref cornstarch	Suspension	No bubbles	Blue-black
Unknown U	Dissolves	Bubbles	No blue-black

F3. Powder test station record (original synthetic teaching figure)

Answer: _____
_____
_____

28. (2 points) Use Figure F3. Which displayed test most directly distinguishes the starch reference through its characteristic color response?

Refer to F3. Powder test station record (original synthetic teaching figure) shown with the first question in this figure group.

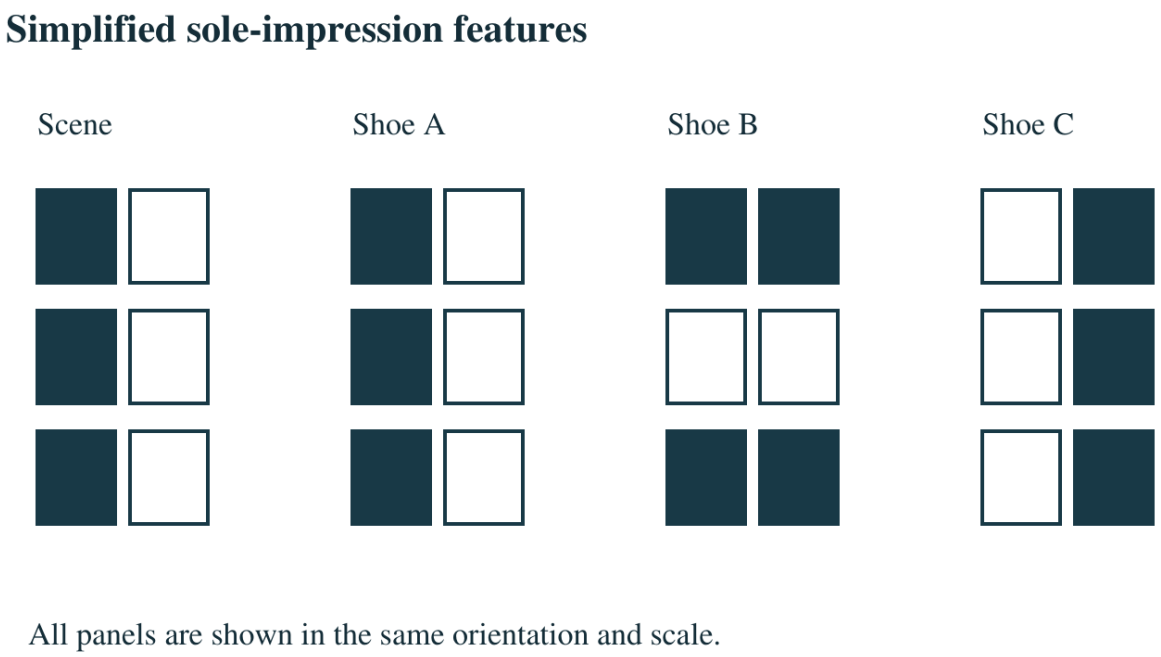
Answer: _____
_____
_____

29. (2 points) Use Figure F3. Why should an identification compare the complete row rather than only water solubility?

Refer to F3. Powder test station record (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

30. (2 points) Use Figure F4. Which candidate shares the scene class pattern in the displayed orientation?



F4. Class-pattern comparison (original synthetic teaching figure)

Answer: _____
_____
_____

31. (2 points) Use Figure F4. How many filled blocks are visible in the scene panel?

Refer to F4. Class-pattern comparison (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

32. (2 points) Use Figure F4. What additional type of feature could help distinguish shoes sharing this class pattern?

Refer to F4. Class-pattern comparison (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____